

Epithelial barrier biology: good fences make good neighbours



The external surfaces of the body, such as the skin and the gastrointestinal mucosal membrane, are an important line of defence preventing the invasion of microorganisms and their products. Mucosal immune cells, especially intraepithelial lymphocytes, are involved in maintaining the integrity of these epithelial barriers. They contribute towards the tolerance to commensal organisms, which occupy these same sites, and to the immune responses against harmful organisms and their products. The composition of the microbiota is influenced by immune cells as well as external environmental factors, especially the use of antibiotics and diet. There is an increasing appreciation that the microbiota affects systemic immune responses in addition to local immunity. Failure to control the occupancy by microorganisms may result in the disruption of the delicate homeostasis between beneficial and harmful microorganisms and contribute to inflammatory pathologies. This review will discuss some of our current understanding of the impact of immune cells and diet on the microbiota.